

Course milestone M16: Reproducible Package and the Peer Cold Run

HONR 46400 · Evidence-Driven Research

What to submit

One file, `lastname_m16.pdf`. Everything this milestone asks for goes inside it, as sections in the order below.

Submit it on Brightspace, under **Assignments** then **M16**. Brightspace carries the deadline.

This milestone presents **Book Milestone 11: Your reproducible package** (checkpoint, version 1) and **Book Milestone 9: Your bounded claims** (revisit, version 2 - the research note after public criticism) of the course book, EDR|AI.

Milestone 11: Your reproducible package

Open the milestone workbook in Colab

This chapter closes Studio 11: Reproduce and package. Its lessons each ended at **It is your turn**; what you wrote there arrives here as working material, and this is where the pieces become one artifact you can defend.

What this milestone produces. An independent cold-run record and a reusable research package. Working alone, the record is a clean-environment rerun labeled “solo proxy; external cold run pending”: the label, and the honesty it enforces, are part of the artifact.

What you bring

You also carry forward the last milestone’s artifact: **Milestone 10: Your artifact, ready to publish or present**. This one builds on it, and the chain assumes it is versioned and filed.

Check you are carrying each of these before you start. If one is missing, go back and work that lesson’s **It is your turn** first; the practice below assumes it.

- Lesson 36: Replication and Reproduction: your cold reproduction record: every number re-run from the top, one undisclosed choice varied, and missingness classified before any filling.
- Lesson 37: Open and Reusable Research Packages: your research capsule: notebook, data or its access recipe, environment record, README, licence with its reuse boundary and restrictions, the exact output locations, and a cold rerun in a second environment.

The practice

1. Assemble the draft package first: data or its access route, code, environment, documentation, and licence.
2. Run your own package cold, from a clean environment, following only your written instructions.
3. Then the external test: have someone else run it without your help, and log every place they had to ask you a question. Working alone? Rerun it yourself after a real break, instructions only, and label the record ‘solo proxy; external cold run pending’: a proxy never becomes a claim of independent reproduction.
4. Fix what the runs exposed: every question asked is a missing line in your documentation, and freeze the final manifest with its date.

A version, not a pass

Your milestone artifact is a dated, numbered **version** with the reason for the version attached. When later evidence changes it, you write the next version rather than editing the last one, because the sequence of changes is itself part of your research record.

The four rails, here

Four concerns cross every studio in this book. At this milestone they take these specific forms.

- **Ethics, permissions, and data exposure.** What can be shared is set by your permissions, and the package says so explicitly.
- **Evidence, provenance, and reproducibility.** Reproduction is the evidence rail's final test.
- **AI activity, verification, and human decisions.** An assistant can generate documentation; only a cold human run can validate it.
- **Uncertainty, claim boundary, and revision history.** Reproduce the uncertainty statement, not only the point estimate.

Where this milestone sits

You arrived carrying Milestone 10: Your artifact, ready to publish or present. Milestone 12 decides whether the packaged work is ready to leave your hands, and what the next cycle should ask. The chain ends at Milestone 12: Your release and next cycle, where you decide whether your finished research artifact leaves your hands.

What sends you back

A failed cold run is a finding about your work, and it precedes any release.

How this milestone is assessed

Each row scores **0, 1, or 2. 10 points total.**

#	Criterion	0	1	2
1	The milestone artifact exists in full, specific to your own project	absent, or generic text that would fit any project	present but incomplete, or specific in some parts and generic in others	complete and unmistakably about your project
2	The milestone is written as a dated, numbered version with its reason	no version, or a version with no reason given	versioned, but the reason is decorative rather than an account of what changed	versioned with a reason a reader could use to reconstruct your thinking
3	Every judgment in the artifact is defended by you, not asserted by a tool	conclusions appear with no reasoning, or reproduce a tool's wording	reasoning present but thin where the decision was hardest	each judgment carries a reason you could defend under questioning
4	The four rails are carried in the form this studio requires	one or more rails absent where this studio required them	all rails present but one is nominal	every rail addressed in the form this studio requires
5	The milestone's own product is present and defensible: An independent cold-run record (working alone: a clean-environment rerun labeled 'solo proxy; external cold run pending') and a reusable research package	the milestone's own product is missing	present but would not survive a careful question	present and defensible as stated

! Blocking gate

No work at or after Studio 4 proceeds past a not-authorized determination. This is not scored and cannot be averaged away.

Your workbook

Open the workbook with the link above. It walks these steps with a cell for each one, ends by writing your milestone version with its reason, and adds the rows this studio contributes to your AI Research Ledger and your Research Dossier.

Milestone 9: Your bounded claims

Open the milestone workbook in Colab

This chapter closes Studio 9: Write, bound, and disclose. Its lessons each ended at **It is your turn**; what you wrote there arrives here as working material, and this is where the pieces become one artifact you can defend.

What this milestone produces. A stand-alone research note or report draft, a claim-to-evidence table, and your AI disclosure statement.

What you bring

You also carry forward the last milestone's artifact: **Milestone 8: Your robustness audit**. This one builds on it, and the chain assumes it is versioned and filed.

Check you are carrying each of these before you start. If one is missing, go back and work that lesson's **It is your turn** first; the practice below assumes it.

- Lesson 28: From Results to Claims: your claims with verbs downgraded to what the design established, each carrying its boundary line.
- Lesson 29: Claim–Evidence Tables: your claim-evidence table, one row per claim with claim, evidence, trace, and boundary, every number traced to its output.
- Lesson 30: AI Disclosure and Research Integrity: your AI-use disclosure, written from the ledger and naming what you did not verify.
- Lesson 31: From Dossier to Research Note: your stand-alone note draft: the problem-gap-question spine, methods, results with uncertainty, discussion, ordered findings with the headline claim's boundary inside, references, your disclosure placed, and the pointer to the package Studio 11 will freeze.

The practice

1. Write your claim as one sentence, with its boundary attached in the same sentence.
2. Build the claim-to-evidence table: every claim, the output that supports it, and the check that verified it.
3. State your uncertainty and your limitations in the words a reader outside your field would use.
4. Write your AI disclosure from your ledger, not from memory.

A version, not a pass

Your milestone artifact is a dated, numbered **version** with the reason for the version attached. When later evidence changes it, you write the next version rather than editing the last one, because the sequence of changes is itself part of your research record.

The four rails, here

Four concerns cross every studio in this book. At this milestone they take these specific forms.

- **Ethics, permissions, and data exposure.** Disclosure includes what you sent to tools, not only what they produced.
- **Evidence, provenance, and reproducibility.** A claim with no row in the table does not appear in the draft.
- **AI activity, verification, and human decisions.** The sentence that ships is yours, whatever drafted it.
- **Uncertainty, claim boundary, and revision history.** Findings are never communicated as certainties.

Where this milestone sits

You arrived carrying Milestone 8: Your robustness audit. Milestone 10 adapts these bounded claims to a real audience and rehearses their defense. The chain ends at Milestone 12: Your release and next cycle, where you decide whether your finished research artifact leaves your hands.

What sends you back

A claim you cannot fit in the table is a claim to cut or a check to run, and both send you back a studio.

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